

Final Project Report

Online Learning Engagement and Student Performance Analysis

Project Name	Online Learning Engagement and Student Performance Analysis
Team ID	Master of Computer Applications
Date	21 September 2026
Tools Used	Microsoft Power BI Desktop, Power Query, DAX, SQL

1. Introduction

1.1. Project Overviews

Online learning platforms record every click, assessment and enrolment decision a student makes, yet institutions often struggle to turn this data into timely action. The project **Online Learning Engagement and Student Performance Analysis** uses Microsoft Power BI to convert student-level records into an interactive dashboard and report that show how engagement, prior education, region and risk level relate to academic results and dropout.

The analysis is based on the **online_education_dataset.csv** file, which holds **32,593 student records across 14 attributes**: demographics (gender, region, highest_education, sideband), academic load (studied credits), platform activity (total clicks), outcomes (engiscope, final_result, pass_flag, dropout_flag) and behavioural indicators (engagement_level, performance_level, risk_level). The data was profiled, cleaned and transformed in Power Query, modelled as a star schema and visualised in Power BI. The final solution tracks five KPIs: **Total Students, Average Score, Average Clicks, Dropout Rate and Pass Rate**.

Domain	Education analytics / online learning
Dataset	online_education_dataset.csv (32,593 records, 14 attributes, ~17.3 MB)

Analysis population	22,422 students (as shown on the dashboard KPI card)
Deliverables	KPI card page, interactive dashboard, two analytical report pages
Key Outcomes	Pass Rate 57.64%, Dropout Rate 20%, Average Score 73, Average Clicks 1.73K

1.2. Objectives

- Analyse the relationship between online engagement (VLE clicks) and academic performance.
- Identify the factors associated with student success and failure, such as prior education, region, gender and studied credits.
- Analyse dropout patterns and identify at-risk students early so that support can be targeted.
- Prepare a clean, reliable dataset by resolving missing values, duplicates and data-type issues.
- Build an interactive, visually consistent Power BI dashboard and report with KPI cards, slicers and drill-down visuals.
- Provide actionable, data-driven recommendations for retention and intervention strategies.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

The Customer Problem Statement template was used to understand the problem from the point of view of the institution or platform administrator.

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
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PS-1	An educational institution or online learning platform administrator	Understand overall student engagement and academic performance and identify factors associated with student success and failure	It is difficult to determine how learning activity, assessment performance, educational background, region and other student characteristics are associated with academic outcomes	Student-level learning data contains many different engagement and academic attributes that need to be analysed together	Uncertain about which factors are associated with better or poorer academic outcomes
PS-2	An educational institution or online learning platform administrator	Analyse dropout patterns and identify students who may be at risk of dropping out	It is challenging to monitor dropout rates, pass rates, engagement levels, and differences among student groups	Dropout behaviour may vary with engagement, academic performance, educational background, region and other student characteristics	Concerned about identifying at-risk students early and supporting data-driven retention and intervention strategies

2.2. Project Proposal (Proposed Solution)

The proposal outlines the objective, scope, problem, proposed approach and resource requirements for the project.

Project Overview	
Objective	Develop an analytical and predictive framework to monitor student engagement metrics in online learning environments and evaluate their direct impact on academic performance.

Scope	Focuses on processing student interaction logs, tracking online participation rates, evaluating assessment scores, building predictive models for student performance, and visualising insights via an interactive dashboard. Excludes offline traditional classroom activities.
Problem Statement	
Description	In online learning environments, educators struggle to track student engagement in real time, leading to delayed interventions for disengaged or struggling students.
Impact	Unidentified disengagement leads to higher course dropout rates, lower overall academic achievement, and inefficient resource allocation for academic support.
Proposed Solution	
Approach	Utilise machine learning algorithms to analyse behavioural engagement patterns (e.g., login frequency, video watch time, assignment submission timeliness) and predict academic performance outcomes.
Key Features	Real-time engagement tracking, automated early warning alert systems for at-risk students, predictive performance scoring, and actionable dashboard visualisations.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Multi-core CPU / GPU (e.g., 2 x NVIDIA V100 GPUs)

Hardware	Memory	8 GB RAM (minimum)
Hardware	Storage	1 TB SSD for raw logs, models and analytical outputs
Software		
Software	Frameworks	Python web/application framework (e.g., Flask, Streamlit)
Software	Libraries	Data processing and ML libraries (e.g., scikit-learn, pandas, numpy, matplotlib)
Software	Development Environment	IDE and version control (e.g., Jupyter Notebook, VS Code, Git)
Data		
Data	Data Source	Educational LMS interaction dataset (e.g., Kaggle LMS dataset / Open University Analytics Dataset, ~10,000+ student records in CSV/JSON format)

2.3. Initial Project Planning

The work was organised into three sprints using a product backlog. Story points reflect relative effort; the total planned effort is **8 story points** delivered between 15 and 21 September 2026.

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Database Environment Setup	USN-1	As a data analyst, I can initialise the online_education_db and import student telemetry so that analytic data has structured storage.	2	High	Individual	15 Sep 2026	17 Sep 2026

Sprint-1	High-Level Summary Analytics	USN-2	As a data analyst, I want to calculate KPI tallies for total active students, average marks, and passed vs. dropout populations.	1	High	Individual	15 Sep 2026	17 Sep 2026
Sprint-2	Engagement & Risk Segmentation	USN-3	As a data analyst, I want to group student metrics by risk, performance, and engagement layers to locate groups needing intervention.	2	High	Individual	18 Sep 2026	19 Sep 2026
Sprint-2	Behavioral Correlation	USN-4	As a data analyst, I want to map score averages across click bands to evaluate how activity scales with test outcomes.	2	Medium	Individual	18 Sep 2026	19 Sep 2026
Sprint-3	Dashboard Delivery & UI Design	USN-5	As a data analyst, I want the interactive dashboard layers to be cleanly structured using Fluent style guidelines for seamless evaluation.	1	High	Individual	20 Sep 2026	21 Sep 2026

Sprint summary

Sprint	Story Points	Duration	Start	End
Sprint-1	3	3 days	15 Sep 2026	17 Sep 2026
Sprint-2	4	2 days	18 Sep 2026	19 Sep 2026
Sprint-3	1	2 days	20 Sep 2026	21 Sep 2026
Total	8	7 days		

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Data Collection Plan

Section	Description
Project Overview	The objective of this project is to analyse online student demographics, engagement levels (click counts), academic scores and risk indicators to understand final student outcomes (Pass, Withdrawn, Distinction, Fail) and optimise online learning strategies.
Data Collection Plan	Data is retrieved from primary online learning management systems (LMS) and student information databases, capturing academic background, region, clickstream activity, assignment scores and final course completion status.
Raw Data Sources Identified	The primary raw dataset online_education_dataset.csv contains 32,593 individual student records tracking 14 demographic, engagement and performance attributes.

Raw Data Sources

Source Name	Description	Location / URL	Format	Size	Access Permissions
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Dataset 1: Online Education Dataset	Student demographic information (gender, region, highest_education, sideband), engagement metrics (total clicks, engagement_level), evaluation indicators (avg_score, studied_credits, performance_level) and outcome indicators (risk_level, pass_flag, dropout_flag, final_result).	online_education_dataset.csv	CSV	~17.3 MB	Public / Open Access
Dataset 2: Online Education Database	Relational SQL database schema containing queries and aggregations for student counts (28,785 unique students), engagement bands, average scores (72.83) and retention/dropout metrics.	Sql queries.sql	SQL	~2 KB	Internal / Open Access

3.2. Data Quality Report

The data quality report summarises the issues found in the source data, their severity and the planned resolution. All issues were found in **online_education_dataset.csv**.

Data Quality Issue	Details	Severity	Resolution Plan
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Missing values in platform engagement metrics	total clicks, engagement_level and risk_level contain 2,852 missing values each (~8.75% of the dataset).	High	Impute missing total clicks with 0. Set missing engagement_level to "None" / "Inactive" and map missing risk_level to "Unevaluated" based on activity thresholds in Power Query.
Missing performance scores and evaluations	avg_score has 5,866 missing values (~18.0%), and performance_level has 5,898 missing values (~18.1%).	High	Replace missing avg_score with 0 or the course-level median for unassessed students; set performance_level to "Unassessed" / "Not Graded".
Missing socioeconomic data (sideband)	1,111 records (~3.41%) lack Index of Multiple Deprivation (sideband) classification.	Moderate	Impute null sideband entries with a distinct "Unknown" category to preserve record count without biasing the socioeconomic distribution.
Data Quality Issue	Details	Severity	Resolution Plan

Duplicate student records	1,356 complete row duplicates and 3,808 repeated id_student identifiers across course module attempts.	Moderate	Remove exact duplicate rows in Power Query; group or aggregate multiattempt student IDs by module attempt or retain the latest attempt record.
Inconsistent categorical formatting in sideband	Inconsistent string notation (e.g., "10-20" missing the % symbol vs "20-30%").	Low	Standardise text patterns using Power Query string replacement ("1020" to "10-20%") to ensure consistent categorical filtering and reporting.
Incorrect identifier data type	id_student is stored as an integer (int64), leading to autosumming and numeric aggregation in BI tools.	Low	Re-type id_student from Integer to Text/String to prevent accidental aggregation and ensure correct relational modelling.

The resolutions actually applied to the data are documented in Section 3.3.

3.3. Data Exploration and Preprocessing

This step identifies the data sources, assesses quality issues such as missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Record flow: 32,593 raw records; 3,808 duplicate id_student records removed (28,785 unique students, matching the SQL count); the Power BI dashboard and report are built on 22,422 students.

Section	Description
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Data Overview	Dataset scope: online_education_dataset.csv contains 32,593 student records across 14 attributes capturing demographic information (gender, region, highest_education, sideband), academic load (studied_credits), platform activity metrics (total clicks), academic outcome metrics (avg_score, final_result, pass_flag, dropout_flag) and behavioural indicators (engagement_level, performance_level, risk_level).
Data Cleaning	Missing values handled: • Imputed sideband with "Unknown". • Filled missing total clicks with 0 and re-evaluated engagement_level as "None". • Filled missing avg_score with the student median score (or overall mean for unassessed students) and updated missing performance_level to "Unassessed". Duplicates: Removed 3,808 duplicate id_student records to retain unique student instances.
Data Transformation	Power Query operations applied: • Filtering: Excluded test records where studied_credits was less than 30 credits. • Pivoting and dynamic aggregation: Pivoted assessment scores to calculate overall course averages. • Calculated columns: Added the click_per_credit ratio (total clicks / studied_credits) and mapped custom risk scoring metrics.
Section	Description
Data Type Conversion	Schema rectification: • id_student: converted to Text/String to avoid integer aggregation. • pass_flag and dropout_flag: cast to Boolean flags (1/0). • total clicks and avg_score: explicitly cast to Decimal/Float types. • sideband, gender, region: categorical text fields.

Column Splitting and Merging	Structural refinement: • Splitting: Split sideband (e.g., 20-30%) into imd_min_pct (20%) and imd_max_pct (30%) numerical boundary columns. • Merging: Combined gender and highest_education into a consolidated demographic key, Demographic_Segment (e.g., M_HE Qualification).
Data Modeling	Relationships and measures: • Built a Star Schema model linking a central Fact_StudentPerformance table to the dimension tables Dim_Student, Dim_Demographics and Dim_Region. • Set up 1:N single-direction relationships from Dimensions to Fact. • Created DAX measures for aggregated retention, pass rates and dynamic click ratios.
Save Processed Data	Storage and future use: Applied all Power Query M-steps, loaded the cleaned table structure into Power BI Desktop / SQL Database, and exported the finalised cleaned dataset as cleaned_online_education.csv for analytics.

4. Data Visualization

4.1. Framing Business Questions

Well-framed business questions guide the choice of metrics, the selection of data and the design of meaningful, actionable visualisations in Power BI. Eight business questions were defined for this project.

No.	Question	Business Question	Planned Visualization
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1	Overall Final Result Distribution	What is the distribution of students across different final academic results (Pass, Fail, Distinction, Withdrawn)?	Pie / Donut chart showing the count and proportion of students for each final result status.
2	Student Academic Performance Across Regions	How does student academic performance (average score) vary across different geographical regions?	Clustered bar chart / Map visualisation comparing average assessment score across regions.
3	Engagement Level vs Final Result	How does student VLE engagement level (total clicks) influence their likelihood of passing or withdrawing?	Stacked column chart of students by Engagement Level (Low, Medium, High), stacked by Final Result.
4	Academic Performance by Prior Educational Background	Which highest education qualification category yields the highest average scores and completion rates?	Clustered column chart comparing average scores and pass rates across prior education levels.
5	Risk Level Analysis by Socioeconomic Status (IMD Band)	How is student academic risk level distributed across different Index of Multiple Deprivation (IMD) bands?	100% stacked bar chart showing the proportion of risk levels (Low Risk, High Risk, Very High Risk, Safe) across IMD bands.
6	Studied Credits and Dropout Rates	Does taking on a higher number of studied credits increase the risk or rate of student dropout?	Line and stacked column chart plotting students enrolled per credit load against dropout rate %.
No.	Question	Business Question	Planned Visualization

7	Gender-Based Comparison of Performance and Risk	Are there significant differences in average scores and risk levels between male and female students?	Matrix / Clustered bar chart comparing total clicks, average scores and risk distributions by gender.
8	Platform Interaction vs Academic Achievement	What is the relationship between overall student VLE activity (total clicks) and final performance level?	Scatter plot of total clicks (X) against average score (Y), segmented by performance level (High, Medium, Moderate).

4.2. Developing Visualizations

The business questions were translated into Power BI visuals across the dashboard and two report pages. The table below shows where each question is answered; screenshots with insights follow in Sections 5 and 6.

No.	Visualization Developed	Location	Status
1	Donut chart: Total Students by final result; column chart: Total Students by final result	Dashboard; Report Page 1	Implemented
2	Column chart: Average Score by region; column chart: Total Students by region	Report Page 1; Dashboard	Implemented (column chart used in place of map)

No.	Visualization Developed	Location	Status
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3	Decomposition tree: Total Students > Engagement Level > Final Result; column charts of average clicks and average score by engagement level	Report Page 2; Dashboard	Implemented
4	Column charts: Pass Rate and Dropout Rate by highest education	Report Page 2	Implemented
5	Pie chart: Total Students by risk level; decomposition tree: Risk Level > Highest Education > Gender > Region	Report Page 2; Report Page 1	Partially (IMD-band split not yet visualised)
6	Studied credits vs dropout rate	n/a	Planned
7	Stacked bar chart: Total Students by gender and final result; gender slicer	Report Page 2; Dashboard	Partially (scores and clicks by gender not shown)
8	Column charts: average clicks and average score by engagement level; pie chart of average score by performance level	Dashboard	Partially (column charts used in place of scatter plot)

5. Dashboard

5.1. Dashboard Design Screenshot with Insights

The dashboard, **Online Education Analytics Dashboard: Student Performance, Engagement & Dropout Analysis**, was designed for quick, at-a-glance reading. It uses a clear top-to-bottom layout (title, KPI cards, charts), a consistent pink and blue colour theme, six interactive slicers on the right-hand side, cross-filtering between visuals, and a natural-language Q&A box for ad-hoc questions.

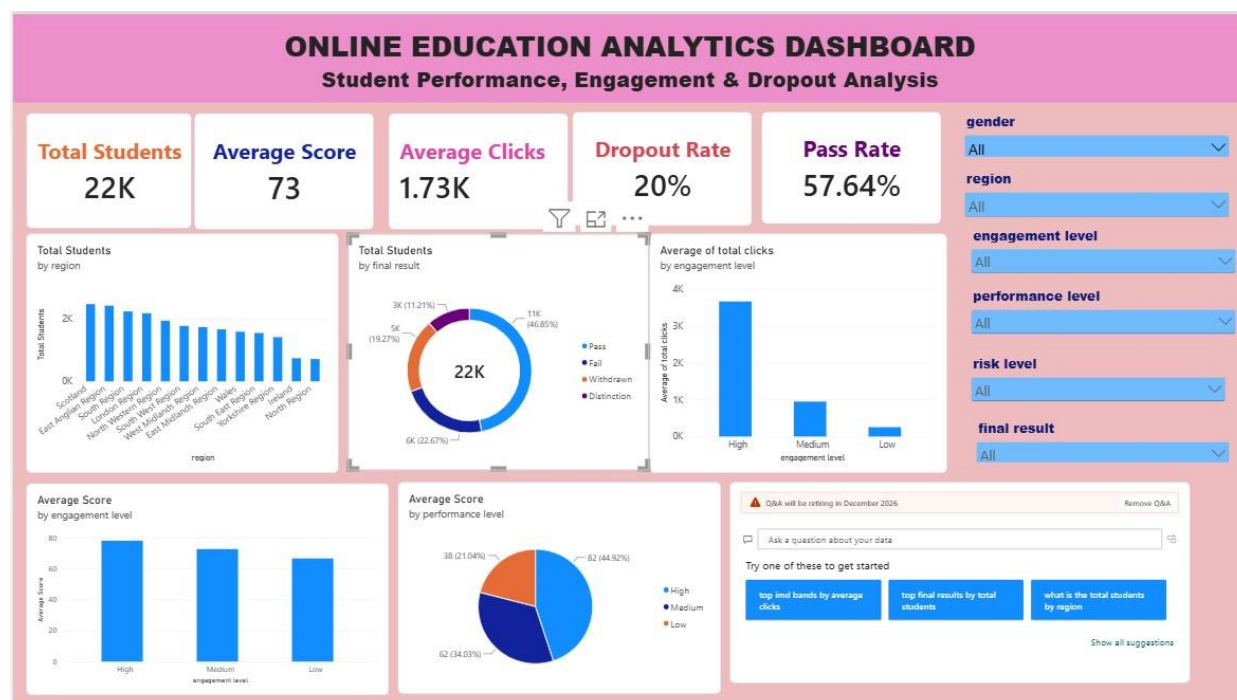


Figure 1: Online Education Analytics Dashboard

Major outcomes from the dashboard

- **KPI snapshot:** The dashboard covers about 22K students (22,422) with an average score of 73, an average of 1.73K clicks per student, a Pass Rate of 57.64% and a Dropout Rate of 20%.
- **Final result mix:** Pass is the largest outcome at 46.85%, followed by Fail (22.67%), Withdrawn (19.27%) and Distinction (11.21%).
- **Regional enrolment:** Scotland has the largest student population (about 2.5K), while Ireland and North Region have the smallest (about 0.7K each).
- **Engagement and activity:** Average clicks fall sharply across engagement bands, from roughly 3.7K for High engagement to about 1K for Medium and well under 0.5K for Low.

- **Engagement and score:** Average score rises with engagement, from roughly 67 (Low) to about 73 (Medium) and 78 (High).
- **Performance levels:** Average scores are 82 for High, 62 for Medium and 38 for Low performance students, a gap of 44 points between the top and bottom groups.
- **Interactivity:** Slicers for gender, region, engagement level, performance level, risk level and final result let users isolate any cohort, and the Q&A box supports questions such as "top IMD bands by average clicks".

6. Report

6.1. Report Design Screenshot with Insights

The report provides a detailed, structured account of student performance, engagement and dropout trends across regions, education levels and risk groups. It is organised into four pages: KPI Cards, Dashboard, Report Page 1 and Report Page 2. The Dashboard page is shown in Section 5.

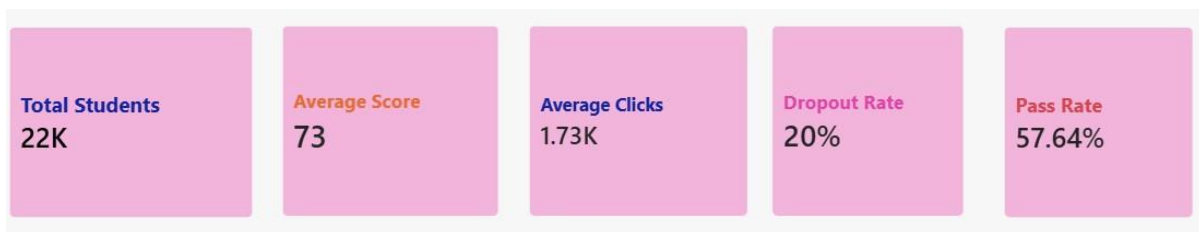


Figure 2: KPI Cards page (Total Students, Average Score, Average Clicks, Dropout Rate, Pass Rate)

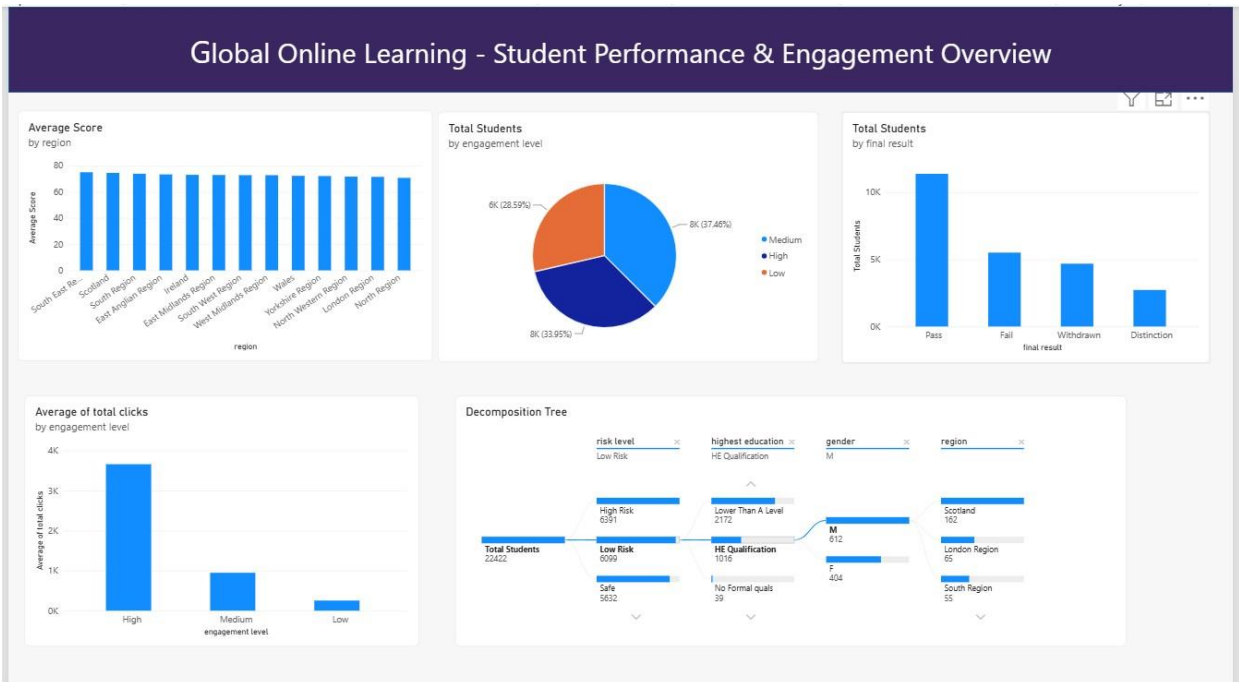


Figure 3: Report Page 1, Global Online Learning: Student Performance & Engagement Overview

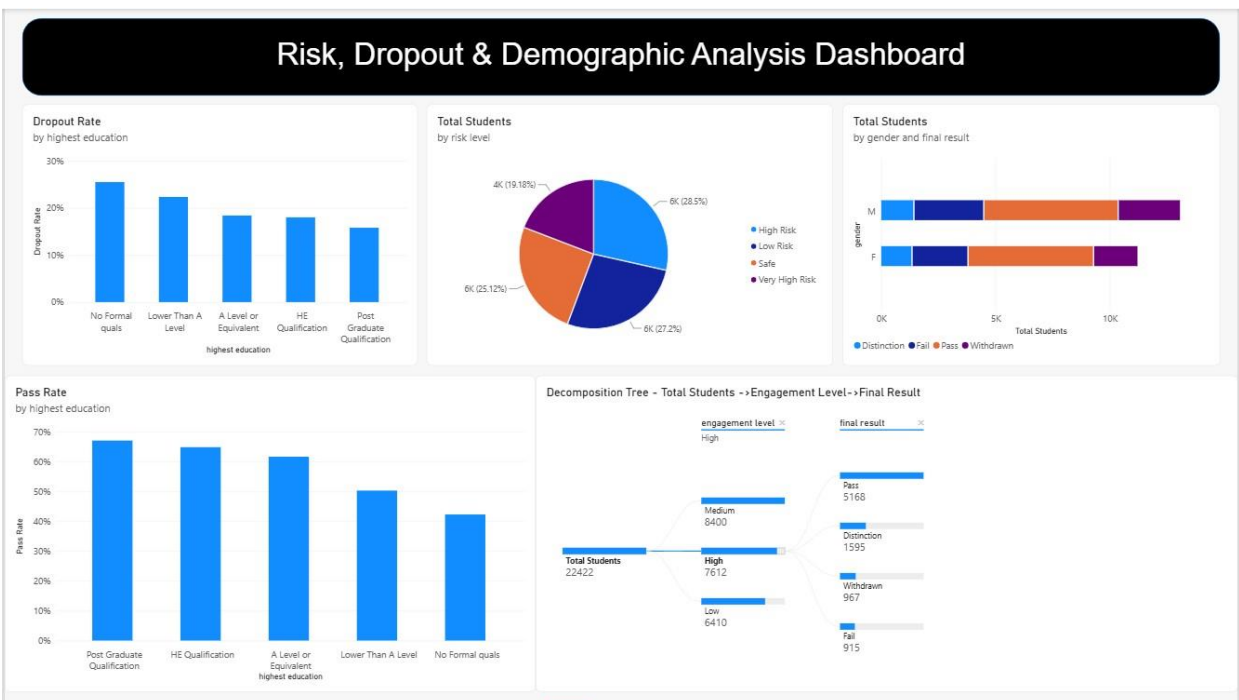


Figure 4: Report Page 2, Risk, Dropout & Demographic Analysis Dashboard
Key insights from the report

1. KPI Overview

- The report covers **22,422** unique students with an average score of **73** and an average of **1.73K** clicks per student. The overall Pass Rate is **57.64%** and the Dropout Rate is **20%**.

2. Engagement vs Performance

- Students are split across Medium (**8,400**; 37.46%), High (**7,612**; 33.95%) and Low (**6,410**; 28.59%) engagement. High-engagement students record the highest average score (about 78), while Low-engagement students score about 67, and average clicks rise from under 0.5K to about 3.7K across the same bands. Higher click activity is associated with better results, so encouraging platform activity should improve outcomes.
- Of the 7,612 High-engagement students, **5,168** passed and **1,595** earned a Distinction, while only **967** withdrew and **915** failed.

3. Dropout Analysis

- Dropout Rate is highest for students with **No Formal quals** (about 25%) and **Lower Than A Level** (about 22%), and lowest for **Post Graduate Qualification** (about 16%). Low engagement and low prior education together act as early-warning signals for dropout.

4. Risk Distribution

- **High Risk** is the largest group (about 6K students, 28.5%; 6,391), followed by Low Risk (27.2%; 6,099), Safe (25.12%; 5,632) and Very High Risk (19.18%; about 4.3K). High Risk and Very High Risk together represent roughly **47.7%** of students, so these groups should be prioritised for support.

5. Regional Disparities

- Scotland has the most students, while Ireland and North Region have the fewest. Average scores are fairly close across regions (roughly 71 to 75), with South East Region leading and North Region trailing, suggesting the need for targeted regional support rather than a large regional performance gap.

6. Education Level Analysis

- Students with a **Post Graduate Qualification** achieve the highest Pass Rate (about 67%), followed by HE Qualification (about 65%) and A Level or

Equivalent (about 62%). **No Formal quals** shows the lowest Pass Rate (about 42%) and the highest Dropout Rate (about 25%). Learners with lower prior education may need extra guidance.

7. Gender and Final Result

- Male students form the larger group. For both genders Pass is the largest final-result segment, with Fail, Withdrawn and Distinction making up the remainder. The decomposition tree on Report Page 1 also lets viewers drill from risk level and education into gender and region (for example, Low Risk students with an HE Qualification: 612 male and 404 female).

8. Recommendations

- Target low-engagement and high-risk students with reminders and mentoring, add support for regions and education levels with high dropout, and track Pass Rate and Dropout Rate monthly against agreed goals.

7. Performance Testing

7.1 Utilization of Data filters

Filters were used to make the dashboard interactive and to keep the data focused on valid student records.

Filter Type	Fields / Location	Purpose
Slicers (6)	gender, region, engagement level, performance level, risk level, final result (Dashboard page)	Let users segment every visual by student characteristics and outcomes.
Decomposition tree drill-down (2)	Report Page 1: Risk Level > Highest Education > Gender > Region. Report Page 2: Engagement Level > Final Result	Allow users to drill from Total Students into deeper segments.
Cross-visual filtering	All visuals on the Dashboard and Report pages	Selecting a bar, slice or segment filters related visuals.

Filter Type	Fields / Location	Purpose
Power Query filter	studied_credits < 30 excluded	Removes test records before modelling.
Q&A naturallanguage query	Dashboard page	Supports ad-hoc questions without building new visuals.

7.2 No of Calculation Columns/Measures

Calculated / derived columns: 5

Column	Type	Definition
click_per_credit	Calculated column	total clicks / studied_credits
imd_min_pct	Split column	Lower boundary of sideband (e.g., 20%)
imd_max_pct	Split column	Upper boundary of sideband (e.g., 30%)
Demographic_Segment	Merged column	gender combined with highest_education (e.g., M_HE Qualification)
Custom risk score	Calculated column	Mapped custom risk scoring metric

DAX measures: 5 KPI measures plus supporting measures

Measure	Purpose
Total Students	Count of unique students (22,422)
Average Score	Average assessment score across students (73)
Average Clicks	Average total VLE clicks per student (1.73K)

Dropout Rate	Percentage of students flagged as dropped out (20%)
Pass Rate	Percentage of students flagged as passed (57.64%)
Measure	Purpose
Retention and click-ratio measures	Supporting DAX measures for aggregated retention and dynamic click ratios in the data model

7.3 No of Visualization

Page	Visuals	Count
KPI Cards	5 KPI cards	5
Dashboard	5 KPI cards, 3 column charts, 1 donut chart, 1 pie chart, 1 Q&A visual, 6 slicers	17
Report Page 1	3 column charts, 1 pie chart, 1 decomposition tree	5
Report Page 2	2 column charts, 1 pie chart, 1 stacked bar chart, 1 decomposition tree	5
Total	15 charts, 10 KPI cards, 6 slicers, 1 Q&A visual	32

8. Conclusion/Observation

The project shows that student engagement, prior education and risk profile are strongly associated with academic outcomes in the online learning data. Across 22,422 students the programme records an average score of 73, a Pass Rate of 57.64% and a Dropout Rate of 20%. The main observations are:

- **Engagement matters:** Average clicks and average scores both rise from Low to High engagement, and High-engagement students are far more likely to pass or earn a Distinction than to withdraw.

- **Prior education shapes outcomes:** Pass Rate ranges from about 42% (No Formal quals) to about 67% (Post Graduate Qualification), and Dropout Rate from about 16% to about 25% across the same groups.
- **A large at-risk population exists:** About 47.7% of students fall into the High Risk or Very High Risk categories, making risk-based targeting a priority.
- **Regional differences are modest:** Average scores lie in a narrow band (roughly 71 to 75), so support should be targeted by engagement and education level as well as region.
- **Data preparation was essential:** Handling missing values, removing 3,808 duplicates and correcting data types produced a reliable, analysis-ready dataset.

Overall, the Power BI dashboard and report give administrators a clear, interactive view of student success, dropout and risk. They support early identification of at-risk students and data-driven retention strategies, meeting the objectives set in Section 1.2.

9. Future Scope

- **Predictive modelling:** Build machine learning models to predict final result and dropout probability, as outlined in the project proposal.
- **Automated early-warning alerts:** Trigger alerts for students whose engagement or risk level crosses defined thresholds.
- **Time-based analysis:** Add dated activity and assessment data to study weekly engagement trends and the timing of dropout.
- **Deeper socioeconomic and workload analysis:** Complete the IMD-band risk analysis and the studied-credits vs dropout analysis planned in Section 4.
- **Near real-time refresh:** Connect Power BI directly to the LMS or SQL database with scheduled refresh in Power BI Service.
- **Role-based access:** Publish to Power BI Service with row-level security so regional managers and advisors see only their own cohorts.
- **Additional data sources:** Include attendance, forum participation and course feedback to enrich the engagement picture.

10. Appendix

10.1. Source Code

The following artefacts were used to prepare and analyse the data.

Artefact	Description
Sql queries.sql	SQL queries on online_education_db: student counts (28,785 unique students), KPI tallies (average score 72.83, passed vs dropout populations), grouping by risk, performance and engagement levels, and an average-score query across click bands (0500, 500-1000, 1000-2000, 2000-3000, 3000+).
Power Query M steps	Missing value imputation, duplicate removal, filtering, data type conversion, column split/merge and the click_per_credit calculation (Section 3.3).
DAX measures	Total Students, Average Score, Average Clicks, Dropout Rate, Pass Rate, and retention/click-ratio measures.
cleaned_online_education.csv	Finalised cleaned dataset exported after preprocessing.

10.2. GitHub & Project Demo Link

GitHub Repository	https://github.com/deepikatelu005-lab/Online-Learning-Engagement-and-Student-Performance-Analysis
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